

# LIVER (Surgery)

## Anatomy

- RUQ, ~1.5 kg. **Middle hepatic vein divides into right (60%) + left lobes.** Glisson's capsule + peritoneum (except posterior "**bare area**").
- Ligaments: left/right triangular (to diaphragm), **falciform (remnant umbilical vein)**, lesser omentum (hilar structures in free edge).
- **8 segments (Couinaud).** Blood: **portal vein 80% + hepatic artery 20%**; 3 hepatic veins (R/M/L). **LPV longer than RPV.** Common hepatic a. → (GDA) → proper hepatic → R + L. **Cantlie's line** (GB fossa → MHV) divides functional right/left.
- **Only organ that regenerates.** Anhepatic survival 24–48h.

## Acute & chronic liver disease

- **Clinical:** malaise, jaundice, confusion/drowsiness, abdominal pain/swelling, N/V, ascites, tremor, bruising, oedema, **fetor hepaticus**. **Most useful sign = flapping tremor (asterixis).**
- **Acute liver failure (ALF):** sudden severe dysfunction + encephalopathy + coagulopathy in previously normal liver. **AASLD: INR >1.5 + any encephalopathy + illness <26 weeks.** Incidence <10/million; **mortality 30–40%**; transplant for some.
- **Chronic liver disease:** 3rd leading premature death cause UK; 90% preventable, 75% present late. Hyperdynamic circulation. **Skin: spider naevi, palmar erythema, leukonychia (white nails); endocrine: hypogonadism, gynaecomastia.** Encephalopathy, ascites, sarcopenia.
- **Prognostic scores: Child-Turcotte-Pugh (CTP) + MELD (INR + bilirubin + creatinine).**
- **Surgical mortality:** CTP-A 10%, CTP-B 30%, **CTP-C 75–80%.** MELD: +1% per point up to 20, +2% above 20.

## Investigation

- **US = first-line** for jaundice + RUQ pain (biliary dilatation, not aetiology).
- **CT:** arterial phase detects small **HCC (arterial supply); haemangioma = late venous enhancement;** inflammatory = rim enhancement; stages/resectability (except peritoneal mets).
- **MRI** superior for focal lesions; **MRCP** = biliary tree (≈ direct cholangiography).
- **PET-CT:** functional (FDG); false positive = inflammation.
- **Biopsy of resectable lesions CONTRAINDICATED** → **tumour seeding** (esp. HCC).

## Liver trauma

- 2nd most injured abdominal organ (after spleen). Blunt (contusion/laceration/avulsion, ± spleen/renal injury, higher mortality) vs penetrating (± chest).
- **Blood loss = major early complication** (liver gets 20% CO). **FAST** for free fluid.
- **Management depends on haemodynamic stability, NOT injury grade:** unstable → laparotomy; **stable** → **non-operative** (most). Peritonitis → surgery. Conservative success → discharge 8–10 days, rescan 6–8 wks. Operative: packing, suture ligation, resectional debridement, hepatectomy.

## Portal hypertension

- Mostly **cirrhosis**; also portal vein occlusion, veno-occlusive, **Budd-Chiari**. Asymptomatic itself → presents decompensated (encephalopathy, ascites, variceal bleed).

- **Variceal bleeding (emergency)**: lower oesophagus commonest; resuscitate (ICU, blood); **vit K + FFP** for coagulopathy; platelets if  $<50,000$ ; **splanchnic vasoconstrictors (terlipressin, octreotide, somatostatin) + prophylactic antibiotics**. Endoscopy confirms (30% non-variceal). **Band ligation** (preferred) or sclerotherapy; balloon tamponade; **TIPSS**.
- **TIPSS**: replaced surgical portocaval shunt; preferred for refractory portal HTN. Complications: liver capsule perforation, **encephalopathy 40%**, stenosis 50% at 1yr. **CI: portal vein occlusion**.
- Surgical shunts (selective/portocaval) — now rare (TIPSS + transplant). **Liver transplant = only treatment of both portal HTN + underlying disease**.
- **Ascites**: **SAAG  $>1.1$  g/dL = portal hypertension**; cytology/culture exclude malignancy/TB.

## Specific chronic liver diseases

- **PSC**: associated with **UC**; **young men 30s**; progresses despite colectomy; **↑cholangiocarcinoma + GB cancer risk**; transplant before malignancy.
- **PBC**: **females**; anti-mitochondrial antibodies; malaise/pruritus; transplant definitive.
- **NAFLD/NASH**: **most prevalent chronic liver disease worldwide** (NAFLD 25%, NASH 1.5–6%); 20% NASH → cirrhosis.
- **Budd-Chiari**: hepatic venous outflow obstruction; **young women**; classic triad **abdominal pain + ascites + hepatomegaly**; hypercoagulable in 75% (AT3/protein C/S); **lifelong anticoagulation**.

## Liver infections

- **Ascending cholangitis**: biliary infection + obstruction. **Charcot's triad = jaundice + rigors + RUQ tenderness**. Organisms: **E. coli (25–50%), Klebsiella, Enterobacter**. US dilated ducts + obstructive LFT + blood culture. **Broad-spectrum antibiotics + rehydration + urgent drainage (ERCP/sphincterotomy/PTC)**. Commonest cause = stones.
- **Pyogenic liver abscess**: commonest; polymicrobial (**Klebsiella, E. coli, Strep milleri**); elderly/diabetic/immunosuppressed; **biliary source commonest (35%)**, portal spread (20%). Aspiration + C&S; metronidazole/clindamycin + 3rd-gen cephalosporin.
- **Hydatid (Echinococcus)**: **albendazole** before + after surgery.
- **Amoebic abscess (Entamoeba histolytica)**: **"anchovy sauce" pus** (sterile, no odour); **metronidazole + diloxanide furoate**. Drainage indications: **left lobe (pericardium rupture risk),  $>5$ cm/superficial, poor response 3–5 days, diagnostic uncertainty**. Rupture → empyema/peritonitis/tamponade.
- **Fasciola hepatica**: watercress; **triclabendazole**.

## Liver tumours

### ### Benign

- **Hepatic adenoma**: **women 25–50, OCP-associated**, solitary; **10% → HCC**; rupture/bleed risk. **Excise if  $>5$ cm** (or stop OCP → may regress).
- **Focal nodular hyperplasia (FNH)**: middle-aged women, no liver disease, **hepatocytes + Kupffer cells**; **NO malignant potential → no treatment**. MRI (liver-specific contrast) distinguishes from mets.
- **Haemangioma**: **most common benign** (5–7%, F:M 5:1), often multiple; incidental; resect only if symptomatic.

### ### Cystic

- **Simple cyst** (5% population, F $>$ M); **polycystic liver** (with adult PKD; **10% cerebral aneurysms**); **cystadenoma** (multilocular, septated; mesenchymal stroma = females, CA19-9 may ↑; premalignant → cystadenocarcinoma).

### ### Malignant

- **HCC**: commonest primary liver cancer; **5th cancer in men**; leading death in cirrhosis; **men 3:1**; 80% Asia/Africa. **Risk**: **HBV (>50%)**, **HCV (17x)**, **aflatoxin**, **alcohol**, **obesity/DM**, **cirrhosis**. Median survival 6–20 mo; <50% at 2yr.
- Staging: **Barcelona (BCLC)**. Only 20–40% resectable. CI: extrahepatic mets, multiple/bilobar, main bile duct/portal vein/IVC thrombus. **Recurrence 80% in 5 yrs**.
- **Transplant = Milan criteria** ( $\leq 5\text{cm}$  single or  $\leq 3$  nodules  $\leq 3\text{cm}$ ; no angioinvasion/extrahepatic) → 75% 4-yr survival.
- If inadequate **future liver remnant (FLR)** → **portal vein embolisation (PVE)**. Resection only CTP-A (minor in B; not C).
- **Intrahepatic cholangiocarcinoma (ICC)**: 20% of cholangiocarcinomas; 2nd to HCC; ↑ in SE Asia (**Clonorchis sinensis**). Subtypes: periductal/mass-forming/intraductal. Aggressive — only 30% resectable; **avoid biopsy**; 5-yr survival 3%.
- **Colorectal liver metastases (CRLM)**: up to 70% of CRC develop mets. **Resection = only curative** (5-yr survival 50%); only 20% candidates (+20–30% after chemo). **R0 margin 1cm gold standard** (R1 similar with modern chemo). **FLR 25% sufficient**. Chemo: **5-FU/folinic + oxaliplatin (50–60%)** ± anti-VEGF/EGFR. **Liver-first approach** for synchronous if liver at risk.

### High-yield one-liners

- **Portal vein 80% + hepatic artery 20%**; 8 segments; Cantlie's line; only organ that regenerates.
- **ALF = INR >1.5 + encephalopathy + <26wk**, no prior liver disease; flapping tremor = best sign.
- **CTP + MELD (INR/bilirubin/creatinine)** for prognosis; CTP-C surgery 75–80% mortality.
- **Liver trauma**: management by haemodynamic stability not grade; stable → conservative.
- **Variceal bleed**: terlipressin/octreotide + antibiotics + band ligation → **TIPSS**. SAAG >1.1 = portal HTN.
- **PSC = young men + UC + cholangiocarcinoma**; **PBC = women + AMA**; Budd-Chiari = young women, triad, anticoagulate.
- **Charcot's triad = ascending cholangitis (E. coli)**; amoebic = anchovy sauce pus.
- **Adenoma = OCP**, 10% HCC, excise >5cm; **FNH = no treatment**; haemangioma = commonest benign.
- **HCC: HBV/HCV/cirrhosis**; BCLC staging; Milan criteria for transplant; biopsy contraindicated (seeding).
- **CRC mets**: resection only cure (50% 5yr); FLR 25%; PVE if small remnant.

# LIVER (Surgery) — Revision Table

Bold = high-yield.

## ANATOMY & ASSESSMENT

| Topic            | Key points                                                                                                                                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Anatomy          | <b>MHV divides R (60%)/L lobes; 8 segments; Cantlie's line (GB fossa→MHV)</b> ; falciform = umbilical vein remnant; bare area posterior    |
| Blood supply     | <b>Portal vein 80% + hepatic artery 20%</b> ; 3 hepatic veins; LPV > RPV; <b>only organ that regenerates</b>                               |
| <b>ALF</b>       | <b>INR &gt;1.5 + encephalopathy + &lt;26 wk, no prior disease</b> ; mortality 30–40%                                                       |
| Chronic signs    | Spider naevi, palmar erythema, leukonychia, gynaecomastia; <b>flapping tremor = best sign</b>                                              |
| <b>Prognosis</b> | <b>CTP (A 10%, B 30%, C 75–80% surgical mortality)</b> ; <b>MELD = INR + bilirubin + creatinine</b>                                        |
| Investigation    | <b>US first-line</b> ; CT (HCC arterial, haemangioma late venous); MRCP biliary; <b>biopsy resectable lesion CONTRAINDICATED (seeding)</b> |

## TRAUMA & PORTAL HYPERTENSION

| Topic                 | Key points                                                                                                                                                        |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Liver trauma</b>   | 2nd to spleen; blood loss main complication; <b>FAST</b> ; <b>manage by haemodynamic stability NOT grade</b> — stable → conservative                              |
| Portal HTN            | Mostly cirrhosis; presents decompensated (encephalopathy/ascites/varices)                                                                                         |
| <b>Variceal bleed</b> | Resuscitate + vit K/FFP + platelets <50k + <b>terlipressin/octreotide/somatostatin + prophylactic antibiotics</b> ; <b>band ligation</b> → balloon → <b>TIPSS</b> |
| TIPSS                 | Refractory portal HTN; encephalopathy 40%, stenosis 50%/yr; <b>CI = portal vein occlusion</b>                                                                     |
| Ascites               | <b>SAAG &gt;1.1 g/dL = portal hypertension</b>                                                                                                                    |

## SPECIFIC DISEASES & INFECTIONS

| Condition                    | Key points                                                                                                                    |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>PSC</b>                   | <b>Young men + UC</b> ; ↑cholangiocarcinoma/GB cancer; transplant before malignancy                                           |
| PBC                          | <b>Females</b> ; anti-mitochondrial Ab; pruritus; transplant                                                                  |
| NAFLD/NASH                   | Most prevalent chronic liver disease (NAFLD 25%); 20% NASH → cirrhosis                                                        |
| <b>Budd-Chiari</b>           | Hepatic outflow obstruction; <b>young women</b> ; <b>triad pain + ascites + hepatomegaly</b> ; hypercoagulable; anticoagulate |
| <b>Ascending cholangitis</b> | <b>Charcot's triad (jaundice + rigors + RUQ)</b> ; E. coli; antibiotics + urgent drainage (ERCP)                              |
| Pyogenic abscess             | Klebsiella/E. coli/Strep milleri; biliary source 35%; metronidazole + cephalosporin                                           |
| <b>Amoebic abscess</b>       | E. histolytica; <b>anchovy sauce pus</b> ; metronidazole + diloxanide; drain if left lobe/>5cm/poor response                  |
| Hydatid / Fasciola           | Albendazole (pre+post surgery) / triclabendazole                                                                              |

## TUMOURS

| Tumour                 | Key points                                                                                                                                                                                                           |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hepatic adenoma</b> | <b>Women + OCP</b> ; <b>10% → HCC</b> ; <b>excise &gt;5cm</b>                                                                                                                                                        |
| FNH                    | Middle-aged women; hepatocytes + Kupffer; <b>no malignant potential, no treatment</b>                                                                                                                                |
| Haemangioma            | <b>Most common benign</b> (F:M 5:1); resect only if symptomatic                                                                                                                                                      |
| Cysts                  | Simple (5%); polycystic (PKD, <b>10% cerebral aneurysm</b> ); cystadenoma (premalignant)                                                                                                                             |
| <b>HCC</b>             | Commonest primary; <b>HBV (&gt;50%)/HCV/cirrhosis/aflatoxin</b> ; <b>BCLC staging</b> ; resectable 20–40%; recurrence 80%/5yr; <b>transplant = Milan criteria</b> (≤5cm or ≤3 nodules ≤3cm); <b>PVE if small FLR</b> |
| ICC                    | 2nd to HCC; <b>Clonorchis (SE Asia)</b> ; aggressive, 30% resectable, avoid biopsy, 5-yr 3%                                                                                                                          |
| <b>CRC mets</b>        | 70% of CRC; <b>resection only cure (50% 5yr)</b> ; 20% candidates; <b>R0 1cm margin, FLR 25%</b> ; 5-FU/oxaliplatin + anti-VEGF/EGFR; liver-first approach                                                           |